How Deep Is Your Data? Estimating Recursive Generation Depth via Statistical Fingerprints

Anonymous ACL submission

Abstract

As large language models increasingly train on web text containing their own outputs, a recursive model→data→model loop arises—yet no method estimates how many such iterations a text has undergone. We formalize *recursive generation depth estimation* as ordinal classification and construct a benchmark of 180K passages spanning 3 model families, 3 decoding strategies, and 4 generation depths. A Random Forest over 19 perplexity-based features achieves 70–85% four-class accuracy (chance = 25%), with perplexity tail statistics emerging as consistent depth markers; a parsimonious subset of 5 features retains 97.6% of full performance. We define the *Effective Discriminable Depth* (EDD) metric, finding $\mathrm{EDD}_{0.75}{=}4$ for the best-characterized family, and show that cross-model transfer reaches 66.4% (2.7× chance). Systematic generalization across seven axes—including cross-scale (7B models, 70.2%), cross-domain, and LoRA-rank ablation—confirms that depth signal is robust and resides in distributional statistics inaccessible from surface text (zero-shot LLMs: 30–36%, 42 pp below RF).

1 Introduction

The text on which large language models are trained increasingly contains their own outputs. As synthetic content proliferates across the web, a recursive loop emerges: models trained on web corpora ingest text generated by earlier models, and their outputs feed into the next generation's training data (Shumailov et al., 2024; Alemohammad et al., 2024). This model→data→model cycle is no longer hypothetical—it is already occurring in the wild—yet we lack the ability to determine how many recursive iterations a given text has undergone, a distinction that matters because first-generation synthetic text may be benign or even beneficial for training, while deeply recursive text drives the distributional collapse that degrades model quality (Dohmatob et al., 2024b).

Existing tools address adjacent but distinct problems. Binary detectors such as Binoculars (Hans et al., 2024), Fast-DetectGPT (Bao et al., 2024), and DivEye (Basani and Chen, 2026) classify text as human-written or machine-generated, collapsing all depths $d \geq 1$ into a single category. Watermarking (Kirchenbauer et al., 2023) requires proactive embedding and cannot be applied retroactively. The model collapse literature (Shumailov et al., 2024; Dohmatob et al., 2024b) assumes depth is known a priori; Drayson et al. (2025) filter synthetic data via binary detection without distinguishing depths. No existing method estimates the ordinal recursive generation depth of a text passage from its content alone.

Our key insight is that recursive generation leaves quantifiable statistical fingerprints in text, and these fingerprints are **consistent in direction but scale-dependent in magnitude** across the three model families and three decoding strategies studied. Specifically, perplexity tail statistics—the 90th-percentile perplexity (P90_PPL) and surprisal variance (VAR_SURPRISAL)—emerge as dominant depth indicators, ranking in the top two features by importance in 9/9 and 8/9 model–strategy configurations, respectively. Surprisingly, a base-model control shows non-monotonic AUC decay while its instruction-tuned counterpart is monotonic, suggesting that instruction tuning regularizes degradation dynamics.

We formalize recursive generation depth estimation as ordinal classification and introduce a lightweight pipeline: compute 19 reference-model perplexity features (effective dimensionality ≈4–5 due to multicollinearity; §4.2) and classify with a Random Forest into $K{+}1$ depth classes. To characterize how recursive generation reshapes these features, we construct the *Depth Degradation Atlas*: a 3×3 grid of model families × decoding

strategies spanning 4 depths (180K samples). The atlas reveals strategy-dependent degradation trajectories and we introduce the *Effective Discriminable Depth* (EDD) metric to quantify how far into the recursion chain adjacent depths remain distinguishable.

Our contributions are:

- **Task formalization and benchmark.** We define recursive generation depth estimation as ordinal classification, introduce the EDD metric, and construct a 180K-sample benchmark spanning 3 model families $\times$ 3 decoding strategies $\times$ 4 depths.
- **Depth Degradation Atlas.** A systematic 3×3 characterization reveals that perplexity tail statistics dominate depth discrimination (top-2 in 9/9 and 8/9 cells), decoding strategy reshapes degradation trajectories qualitatively, and $\text{EDD}_{0.75}{=}4$ for the best-characterized family (Pythia, $p{=}0.001$; consistent across all three families).
- **Generalization across seven axes.** Systematic evaluation—cross-model transfer (66.4%), cross-model recursive chains (84.75%), cross-scale (two 7B models), cross-domain, temperature, reference-model, and LoRA-rank—demonstrates that depth signal is robust and not an artifact of any single configuration. A surface-text ablation (zero-shot LLMs at 30–36%, 42 pp below RF) confirms the signal resides in distributional statistics inaccessible from text alone.

2 Related Work

Model Collapse Theory. Shumailov et al. (2024) demonstrate that recursive training on model outputs erodes distributional tails; Dohmatob et al. (2024b) formalize this via scaling laws (Seddik et al., 2024), Dohmatob et al. (2025) establish strong collapse guarantees, and the MAD framework (Alemohammad et al., 2024) extends the analysis to self-consuming models. Subsequent work addresses mitigation (Gerstgrasser et al., 2024; Drayson et al., 2025; Yi et al., 2026; Falahati et al., 2026), non-monotonic dynamics (Bakshi and Chakraborty, 2026), convergence (Marchi et al., 2024; Wang et al., 2026; Keisha et al., 2025; Dohmatob et al., 2024a), and diversity decline (Guo et al., 2024; Briesch et al., 2024; Padmakumar and He, 2024; Gu et al., 2026; Schaffelder and Gatt, 2026). Schaeffer et al. (2025) argue that "model collapse" conflates multiple distinct phenomena, underscoring the need for operationally precise metrics. Dohmatob et al. predict that recursive training erodes distributional tails as a theoretical scaling law (Dohmatob et al., 2024b); we operationalize this prediction into a deployable ordinal classifier, revealing strategy-dependent trajectories and non-monotonic 7B behavior that the theory does not predict. All assume depth is known a priori; we address the inverse problem of estimating depth from text alone.

AI Text Detection. The detection literature (Wu et al., 2025) spans statistical methods—DetectGPT (Mitchell et al., 2023), Fast-DetectGPT (Bao et al., 2024), Binoculars (Hans et al., 2024), DivEye (Basani and Chen, 2026), GLTR (Gehrmann et al., 2019), Ghostbuster (Verma et al., 2024), DNA-GPT (Yang et al., 2024)—and watermarking (Kirchenbauer et al., 2023), with known limits from paraphrasing (Sadasivan et al., 2023), distribution shift (Pudasaini et al., 2026), and generator barriers (Pu et al., 2026). These methods perform binary classification, discarding ordinal depth information; our approach outputs ordinal depth labels, complementing binary detection.

Adjacent Tasks. EditLens (Thai et al., 2026) estimates editing extent but does not model recursive chains; Uchendu et al. (2020) classify *which* model generated text; LOKI (Ye et al., 2025) benchmarks synthetic data detection across modalities. Mohamed et al. (2025) study iterative generation as a "broken telephone" without classifying iteration depth; self-play methods like SPIN (Chen et al., 2024) instead exploit iterative training for improvement. Our task inverts the problem—estimating ordinal depth from text alone, discriminating adjacent depths rather than detecting aggregate degradation.

Ordinal Classification. Frank and Hall (2001) reduce ordinal problems to nested binary tasks and Rennie and Srebro (2005) formalize loss functions for discrete ordered labels. CORAL (Cao et al., 2020) enforces rank consistency via ordered binary tasks; CLOC (Pitawela et al., 2025) extends contrastive learning with multi-margin ordinal losses. We adopt both as baselines (§5) to confirm the signal is accessible to principled ordinal methods.

Decoding Strategy Effects. Decoding strategies shape text statistics in ways that interact with depth signals (Holtzman et al., 2020; Finlayson

et al., 2024; Garces Arias et al., 2025; Wiher et al., 2022; Basu et al., 2021). Our Depth Degradation Atlas (§4) reveals qualitatively different degradation trajectories—temperature sampling accelerates convergence to a low-entropy attractor while top-k attenuates the depth signal—effects beyond single-pass text quality.

3 Method

3.1 Problem Definition

Let f_θ denote a language model and $\mathcal{D}_0$ a corpus of human-written seed text. We define a *recursive generation chain* of depth K as the sequence $\mathcal{D}_0, \mathcal{D}_1, \ldots, \mathcal{D}_K$, where each $\mathcal{D}_k$ ($k \geq 1$) is produced by finetuning f_θ on $\mathcal{D}_{k-1}$ and sampling from the resulting model. Depth $d{=}0$ denotes human-written text; depth $d{=}k$ ($k \geq 1$) denotes text generated by a model trained on depth-$(k{-}1)$ data.

Given a text passage x, the task is to estimate its recursive generation depth $d \in \{0, 1, \ldots, K\}$. This is an ordinal classification problem: depths carry a natural ordering, and $d{=}1$ is structurally closer to $d{=}0$ than $d{=}3$ is, since each generation step compounds distributional drift.

3.2 Multi-Generational Benchmark Construction

We construct a benchmark spanning three axes: model family, decoding strategy, and generation depth.

Model families. We select three 1–1.5B-parameter models with diverse pretraining corpora and architectures: Pythia-1.4B (EleutherAI), OLMo-1B (AI2), and GPT-2 XL (OpenAI, 1.5B). Restricting to a narrow parameter range isolates the effect of architecture and training data from scale.

Decoding strategies. Each model is paired with three decoding strategies: nucleus sampling ($p{=}0.95$) (Holtzman et al., 2020), temperature sampling ($\tau{=}0.9$), and top-k sampling ($k{=}50$). These span the dominant modes of open-ended generation and induce qualitatively different entropy profiles. Our findings on feature importance and discriminability are limited to this grid; transfer to untested models, scales, or strategies is not established.

Depth and scale. The primary benchmark covers depths d0–d3 across all $3 \times 3 = 9$ model–strategy cells, with 5,000 samples per depth per cell (180K samples total). For depth extension experiments (§4), we extend one pipeline (Pythia-1.4B, nucleus) to d0–d5.

Generation pipeline. Starting from a Wikipedia subset as seed text ($\mathcal{D}_0$), each generation cycle proceeds in three steps: (1) LoRA finetuning (Hu et al., 2022) on $\mathcal{D}_{k-1}$, freezing base model weights and training low-rank update matrices (<1% additional parameters; $r{=}16$, $\alpha{=}32$, lr $2{\times}10^{-4}$, batch size 8, 3 epochs, dropout 0.05, architecture-specific targets); (2) sampling 5,000 passages from the LoRA-finetuned model (max 512 tokens, BOS-seeded, no prompt conditioning) under the specified decoding strategy; (3) the generated corpus becomes $\mathcal{D}_k$ for the next cycle. Each depth is produced by a model explicitly trained on the prior depth's data—not by sampling from the unmodified base model. Our BOS-seeded generation may amplify depth signal relative to prompted settings; extending to prompt-conditioned generation is future work. There is no per-passage lineage across depths, ensuring cross-validation splits do not introduce information leakage.

Reference model. All statistical features (§3.3) are computed using Pythia-1.4B as fixed reference model, the smallest generator whose training corpus partially overlaps with the others. For Pythia this constitutes self-model measurement, yet its accuracy (78.2%) does not exceed GPT-2 XL (85.4%), indicating no systematic self-model advantage. Ablations with OLMo-1B and GPT-2 XL as alternative references yield 67–79% accuracy (Appendix A.16), confirming reference model robustness; further discussion in Limitations.

3.3 Statistical Feature Design

We extract a feature vector $\mathbf{f}(x)$ comprising 19 candidate features from the reference-model perplexity distribution over each passage x; ablation (§4.2) shows the top 5 achieve 97.6% of full-set performance, and variance inflation analysis reveals VIF>10 for 14–16 features, placing the effective independent dimensionality at ≈4–5. The features fall into four categories (Table 1).

Perplexity statistics (9 features) capture the per-token perplexity distribution via moments (mean, variance, skewness, kurtosis) and quantiles (p10–p90); tail quantiles prove especially informative as recursive finetuning erodes the high-perplexity tail (§4). *Surprisal statistics* (3 features): mean, variance, and entropy of $-\log p(\text{token})$; surprisal

Table 1: The 19 statistical features comprising $\mathbf{f}(x)$, grouped by category. All perplexity and surprisal features are computed from the reference model (Pythia-1.4B).

Category	Features	n
Perplexity statistics	mean, var, skewness, kurtosis, p10, p25, p50, p75, p90	9
Surprisal statistics	mean, var, entropy	3
Lexical diversity	type–token ratio, hapax ratio, bigram entropy, trigram entropy	4
Repetition	repeated 2/3/4-gram fractions	3

variance captures second-order structure not redundant with perplexity moments. *Lexical diversity* (4 features): type–token ratio, hapax ratio, bigram/trigram entropy. *Repetition* (3 features): repeated n-gram fractions for $n \in \{2, 3, 4\}$.

3.4 Effective Discriminable Depth

Aggregate multi-class accuracy obscures whether discriminability degrades at specific depth boundaries. We introduce the *Effective Discriminable Depth* (EDD) to formalize the frontier beyond which adjacent depths become statistically indistinguishable.

Pairwise discriminability. Let $\mathcal{D} = \{0, 1, \ldots, K\}$ denote the depth set and $T \in (0.5, 1.0]$ a discrimination threshold. For each adjacent pair $(d, d{+}1)$, we define the pairwise discriminability $\mathrm{AUC}(d, d{+}1)$ as the area under the ROC curve of a dedicated binary Random Forest classifier (500 trees, $\lfloor\sqrt{19}\rfloor$ candidate features per split) trained on class-balanced data from $\mathbf{f}(x)$ (§3.3) to distinguish depth-d from depth-$(d{+}1)$ samples, evaluated via 5-fold stratified cross-validation. Each pair uses an independent classifier, avoiding the decision boundary dilution that arises in multi-class settings (§4.4).

Definition 1 (Effective Discriminable Depth)**.** The *Effective Discriminable Depth* at threshold T is:

$$\mathrm{EDD}_T = \max\{d^*{+}1 : \forall\, d' \le d^*,\ \mathrm{AUC}(d', d'{+}1) \ge T\} \quad (1)$$

where $d^* \in \{0, \ldots, K{-}1\}$. If $\mathrm{AUC}(0, 1) < T$, then $\mathrm{EDD}_T = 0$.

EDD_T is the deepest depth d such that every adjacent pair from $(0, 1)$ through $(d{-}1, d)$ remains discriminable at level T. The universal quantifier ensures EDD reports the longest *contiguous* discriminable prefix rather than an isolated high-AUC pair at greater depth.[1] We set $T{=}0.75$ as default; sensitivity at $T \in \{0.65, 0.70, 0.75, 0.80\}$ is reported in §4.

EDD is an operational measure conditional on the classifier and feature set, not an intrinsic data property; however, classifier convergence (Table 5, spread $\le$0.4 pp) implies EDD is invariant across the studied classifier families.

Classifiers. Our primary classifier is a Random Forest (200 trees, unconstrained depth, 5-fold stratified cross-validation). We compare against CORAL ordinal regression (Cao et al., 2020), DivEye (Basani and Chen, 2026), a 2-layer cross-entropy MLP, logistic regression, and a single-feature baseline using mean perplexity alone; results appear in §5.

4 The Depth Degradation Atlas

We present the Depth Degradation Atlas: a systematic empirical map of how statistical fingerprints evolve across recursive generation depths, models, and decoding strategies. The atlas spans $3 \times 3 = 9$ model–strategy cells evaluated at depths $d \in \{0, 1, 2, 3\}$, with depth extensions to $d{=}5$ for cross-model EDD validation (§4.4).

4.1 In-Distribution Performance

Table 2 reports four-class Random Forest accuracy (5-fold CV, 200 trees) across all nine cells. Accuracy ranges from 69.6% (Pythia, top-k) to 85.4% (GPT-2 XL, temperature), all exceeding the 25% chance baseline by a wide margin. Two axis effects emerge. GPT-2 XL achieves the highest accuracy under every strategy (76.6–85.4%), followed by OLMo (74.2–82.0%) and Pythia (69.6–78.2%). Top-k sampling yields the lowest accuracy in every model. The mean within-model strategy gap is 8.4 pp, comparable to the 7.1 pp mean within-strategy model gap—decoding strategy and model family contribute roughly equally to performance variance.

[1] Our AUC profiles are empirically monotonically decreasing for the 1–1.5B base models (§4), so in practice $\mathrm{EDD}_T = \max\{d'{+}1 : \mathrm{AUC}(d', d'{+}1) \ge T\}$. Qwen2.5-7B exhibits a U-shaped AUC pattern rather than monotonic decay, a base-model control shows non-monotonic decay persists without instruction tuning (§5.4). For non-monotonic profiles, the contiguous formulation is more conservative (see Limitations).

Table 2: Four-class RF accuracy (%) across the 3×3 atlas, with d1–d3 three-class accuracy in parentheses (chance = 25% / 33%). Depth-0 recall exceeds 98% in all cells. Bootstrap 95% CI half-widths range 1.3–1.6 pp across all cells.

	Nucleus	Temp. τ=0.9	Top-k
Pythia-1.4B	78.2 (71.2)	76.3 (70.6)	69.6 (58.8)
OLMo-1B	78.8 (75.0)	82.0 (76.3)	74.2 (66.1)
GPT-2 XL	83.3 (79.0)	85.4 (82.6)	76.6 (69.5)

Table 3: Top-5 features by Gini importance. **Bold**: top-2. `p90_ppl` top-2 in 9/9; `var_surprisal` 8/9 under SHAP (7/9 under Gini).

Cell	Top-5 features (Gini importance, ranked)
Pythia, nuc.	**var_surp (.17)**, **p90_ppl (.15)**, m_ppl (.11), var_ppl (.08), p75 (.07)
Pythia, τ	**var_surp (.17)**, **p90_ppl (.14)**, m_ppl (.11), p75 (.07), m_surp (.07)
Pythia, top-k	**var_surp (.16)**, **p90_ppl (.12)**, m_ppl (.11), var_ppl (.07), m_surp (.06)
GPT-2 XL, nuc.	**var_surp (.21)**, **p90_ppl (.21)**, p75 (.10), m_surp (.07), m_ppl (.06)
GPT-2 XL, τ	**p90_ppl (.25)**, **p75 (.15)**, m_surp (.13), var_surp (.12), p50 (.05)
GPT-2 XL, top-k	**p90_ppl (.24)**, **p75 (.12)**, var_surp (.10), m_surp (.09), ttr (.04)
OLMo, nuc.	**p90_ppl (.18)**, **var_surp (.15)**, p75 (.11), m_surp (.07), ttr (.06)
OLMo, τ	**p90_ppl (.21)**, **var_surp (.16)**, p75 (.12), m_surp (.09), ttr (.05)
OLMo, top-k	**p90_ppl (.20)**, **var_surp (.17)**, p75 (.10), m_surp (.07), m_ppl (.05)

Per-class recall distribution. Depth-0 recall exceeds 98% in all 9 cells, confirming that human-written text is trivially separable from any generation depth. Among generated depths, d2 consistently yields the lowest recall (46–73% across cells), followed by d3 (58–82%) and d1 (72–91%). For example, Pythia-1.4B nucleus achieves per-class recall of d0 = 99%, d1 = 82%, d2 = 59%, d3 = 72%. The strongest d2 recall is obtained by GPT-2 XL nucleus (73%). The d2 trough reflects its position at the center of the depth gradient: d2 features overlap with both the d1 distribution above and d3 below, making it the most confusable class—consistent with the d2-vs-d3 pair being the EDD bottleneck (§4.4). Full per-class recall for all cells appears in Appendix A.3.

4.2 Feature Consistency Across the Grid

Despite the 15.8 pp accuracy spread across cells, a small feature family dominates in every setting. Perplexity tail statistics as a feature group dominate depth discrimination: the 90th-percentile perplexity (`p90_ppl`) ranks top-2 by Gini importance in all 9 cells; surprisal variance (`var_surprisal`) in 8/9 under SHAP (7/9 under Gini; Appendix A.2).[2] Table 3 shows the full per-cell ranking.

Cross-model feature-rank Spearman correlation averages $\bar{\rho} = 0.868$ (range $[0.619, 0.954]$): models agree on which features carry depth information even when they differ in absolute accuracy. Among strategies, top-k yields the highest cross-model rank correlation ($\rho = 0.811$)—the deterministic vocabulary truncation imposes a more uniform feature landscape across architectures, at the cost of weaker overall signal (§4.5).

Percentile-based features outperform their moment-based counterparts. Token-level perplexity distributions are heavy-tailed: individual outlier passages can dominate the arithmetic mean of their depth class, whereas `p90_ppl` is robust to such extremes while still capturing the tail erosion that signals generation depth.

Feature ablation confirms this concentration: the top-2 features (`p90_ppl` + `var_surprisal`) alone reach 91.1% of full accuracy (71.5%), and the top-5 features achieve 97.6% (76.5% vs. 78.4%), with perplexity statistics alone reaching 76.6% (vs. surprisal 73.2%, other categories $\leq$47%). Variance inflation analysis reveals VIF$>$10 for 14–16 of 19 features, placing the effective independent dimensionality at $\approx$4–5; consequently, the stable signal unit is the feature *group* (perplexity tail statistics) rather than individual features. Forward selection independently confirms the dominant pair: `p90_ppl` is selected first in 6/9 cells, `var_surprisal` in 3/9.

The feature hierarchy also shifts with depth. At shallow boundaries (d1 vs. d2), `var_surprisal` dominates importance; at deeper boundaries (d3 vs. d4, d4 vs. d5), tail perplexity statistics (`p90_ppl`, `kurtosis_ppl`) take over. This transition aligns with model-collapse dynamics (Shumailov et al., 2024): early recursion narrows the surprisal distribution (captured by its variance), while later steps erode the perplexity tail (captured by upper quantiles and kurtosis).

4.3 Monotonicity and Inter-Depth Distances

Depth features exhibit strong monotonic trends: perplexity tail statistics (`p90_ppl`, `mean_surprisal`) decrease with depth (Spearman $\rho = -1.0$ across cell-level medians) while repetition features increase, reflecting the complementary narrowing and self-repetition of distribution collapse. Inter-depth distances are non-uniform: d0-vs-d1 AUC exceeds 0.99 in all 9 cells, whereas d2-vs-d3 ranges from 0.69

[2] Individual feature rankings should be interpreted cautiously given high multicollinearity (VIF$>$10 for 14–16/19 features); the group-level dominance of perplexity tail statistics is robust across importance measures.

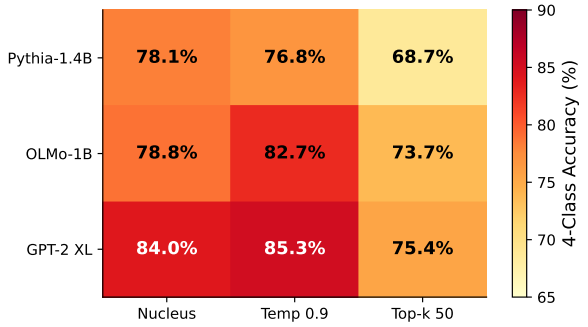


Figure 1: Atlas feature importance heatmap. Rows: 19 features grouped by category (§3.3); columns: 9 model–strategy cells. Color intensity encodes RF Gini importance. The perplexity-tail and surprisal-variance blocks maintain high importance across all cells.

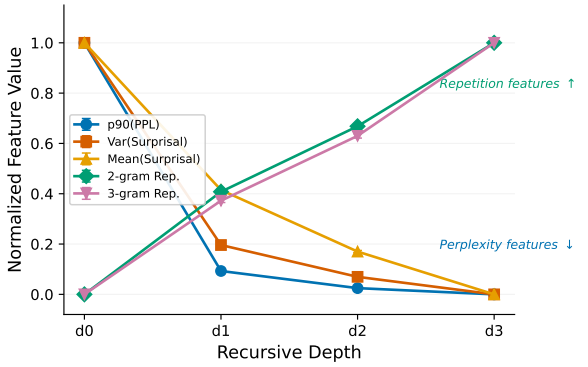


Figure 2: Feature trajectories across depths d=0 to d=3, stratified by model (columns) and strategy (line style). Top row: `p90_ppl` (monotonically decreasing). Middle: `var_surprisal` (decreasing). Bottom: `rep_3gram` (increasing). Shaded bands: interquartile range. The d=0 $\rightarrow$ d=1 gap exceeds all subsequent inter-depth gaps.

to 0.89—separating adjacent deep generations requires the tail statistics identified in §4.2.

4.4 EDD Analysis

We extend the benchmark to depth d=5 for all three models under nucleus sampling. Each model's depth-3 LoRA checkpoint is used to finetune and generate depth-4 data, and the process repeats for depth 5; feature extraction uses the same reference model (Pythia-1.4B). We train dedicated binary RF classifiers (500 trees) for each adjacent pair, as defined in Eq. 1.

AUC decay profile. Figure 3 plots the binary pairwise AUC decay curves. For Pythia-1.4B (nucleus), AUC decays monotonically: d0-vs-d1 = 0.999, d1-vs-d2 = 0.925, d2-vs-d3 = 0.823, d3-vs-d4 = 0.784, d4-vs-d5 = 0.683. The steepest drop occurs between the d1-vs-d2 and d2-vs-d3 pairs; the curve then flattens as the system approaches an indistinguishable regime gradually rather than crossing a sharp boundary.

Table 4: Number of atlas cells achieving EDD $\geq$ 3 at each threshold T, across 9 model–strategy cells (depths 0–3). The d2-vs-d3 boundary is the bottleneck; top-k cells drop first.

	T=0.65	T=0.70	T=0.75	T=0.80
Cells with EDD $\geq$ 3	9/9	8/9	7/9	5/9

Cross-model convergence at depth 4–5. The d3-vs-d4 AUC consistently clusters near the T=0.75 threshold across all three models (Pythia 0.784, OLMo 0.765, GPT-2 XL 0.750), while d4-vs-d5 uniformly falls below 0.70 (0.681–0.684). The directional pattern is consistent, but statistical strength is stratified: bootstrap testing (10,000 resamples, H_0: AUC $\leq$ 0.75) yields p = 0.001 for Pythia (significant), p = 0.091 for OLMo (marginal), and p = 0.505 for GPT-2 XL (non-significant). All three models cross the T=0.75 threshold within the d3–d4 region, establishing a consistent depth-4 decay boundary with graded confidence rather than uniform statistical significance. The d4-vs-d5 saturation is unambiguous across models, supporting the interpretation of d3–d4 as a *transition zone* from discriminable to indistinguishable regime.

EDD threshold sensitivity. Table 4 reports EDD_T across the primary 9-cell benchmark (depths 0–3) at four threshold values. At T=0.65, all 9 cells achieve EDD = 3; at T=0.75, 7 of 9 cells maintain EDD $\geq$ 3; at T=0.80, the count drops to 5 of 9. The cells losing EDD=3 first are consistently the top-k configurations (Pythia top-k at T=0.70, then OLMo top-k at T=0.75), where the d2-vs-d3 pair is the consistent bottleneck across all 9 cells.

Decision boundary dilution. EDD uses binary-pair classifiers to avoid *decision boundary dilution*: in Pythia nucleus, d2 vs. d3 AUC degrades from 0.804 (4-class) to 0.689 (6-class)—a 14.3% relative decline despite identical data—because shared decision boundaries dilute per-pair capacity. Binary classifiers eliminate this confound, isolating true adjacent-pair discriminability.

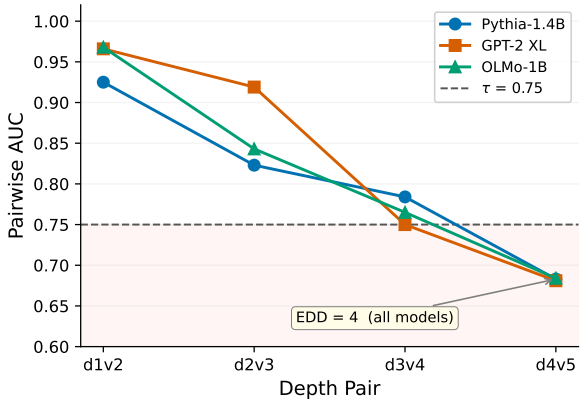


Figure 3: Binary pairwise AUC decay curves for three models (nucleus sampling, depths 0–5). Horizontal dashed lines mark $T \in \{0.65, 0.70, 0.75, 0.80\}$. All three curves converge below 0.70 at d4-vs-d5, while d3-vs-d4 AUC clusters in the range 0.750–0.784. Shaded bands: bootstrap 95% CIs.

4.5 Strategy-Dependent Trajectories

The three decoding strategies induce qualitatively distinct degradation trajectories. Nucleus (p=0.95) produces the most consistent accuracy (78.2–83.3%) and serves as the reference trajectory. Temperature (τ=0.9) yields the highest accuracy in 2/3 models (GPT-2 XL 85.4%, OLMo 82.0%) because uniform logit scaling accelerates convergence to an attractor distribution (Shumailov et al., 2024), amplifying the statistical fingerprint—but this acceleration creates a boundary reversal risk: temperature depth-N features approximate nucleus/top-k depth-$(N{+}1)$ features, causing cross-strategy classifiers to systematically overpredict depth (§5). Top-k (k=50) produces the lowest accuracy (69.6–76.6%) because hard vocabulary truncation removes depth-discriminative low-probability tokens, though feature direction is preserved (cross-model $\rho = 0.811$). The strategy effect (8.4 pp range) exceeds the model effect (7.1 pp), indicating deployment-time knowledge of the decoding strategy matters at least as much as the generator model.

5 Generalization and Robustness

We test whether the atlas results (69.6–85.4%) reflect genuine regularities or training artifacts along seven axes: classifier choice (§5.1), cross-model (§5.2), cross-strategy (§5.3), and cross-scale transfer (§5.4) below; three additional axes in the Appendix.

Table 5: Classifier convergence: 4-class accuracy (%) across three atlas cells on the same 19 features. $\Delta_{\max}$: largest per-cell spread. Chance = 25%.

Cell	RF	XGB	LGBM	$\Delta_{\max}$
Pythia, nuc.	78.8	79.0	78.9	0.2
GPT-2 XL, τ	85.8	86.2	86.2	0.4
Pythia, top-k	68.9	68.9	68.7	0.2

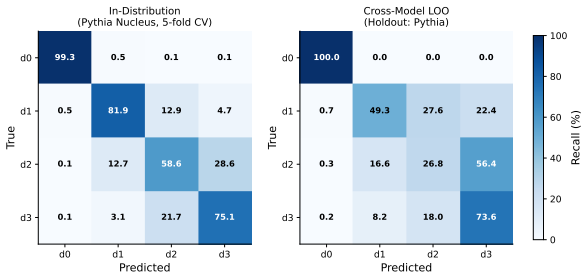


Figure 4: Confusion matrices: best in-distribution (GPT-2 XL, τ=0.9, 85.4%) and best cross-model transfer (OLMo holdout, 69.2%). d0 recall >99% in both; errors spread to d1–d3 under transfer.

5.1 Classifier Comparison and Surface-Text Ablation

Table 5 evaluates RF (bagging), XGBoost, and LightGBM (gradient boosting) on the same 19 features; per-cell spread ≤0.4pp establishes a feature-determined ceiling. On Pythia nucleus, CORAL (Cao et al., 2020) (78.1%) and DivEye (Basani and Chen, 2026) (78.2%) match within-cell RF within 0.1 pp ($p > 0.05$); all feature-based methods outperform CE-MLP (74.9%, $p < 0.001$) and logistic regression (75.0%).

Surface-text ablation. Single-feature regression on mean perplexity performs at chance (25.1%); two zero-shot LLMs (DeepSeek V3.2: 36.2%, Qwen2.5-7B: 30.4%; Appendix A.5) also fail, confirming depth signal resides in distributional statistics inaccessible from surface text (42 pp gap vs. RF).

Convergence across five classifiers confirms the signal resides in features, not the algorithm (Figure 4); errors concentrate at d2/d3 in-distribution and spread to d1–d3 under transfer.

5.2 Cross-Model Transfer

We perform leave-one-model-family-out evaluation (Table 6): train on two families (nucleus), test on the held-out third.

Mean transfer accuracy is 66.4% (2.7× chance); d0 recall (>98%) inflates the headline while d1–d3

Table 6: Cross-model transfer (leave-one-out, nucleus). Chance = 25%/33% (d1–d3). Mean d1–d3 bal. acc: 55.4%. Bootstrap 95% CI half-widths ≤0.4 pp.

Holdout	Acc (%)	d0 Rec.	d1 Rec.	d2 Rec.	d3 Rec.
Pythia	62.4	100.0	49.3	26.8	73.6
OLMo	**69.2**	99.3	75.5	43.3	58.8
GPT-2 XL	67.6	98.5	74.7	52.6	44.5
Mean	66.4	99.3	66.5	40.9	59.0

Table 7: Cross-strategy transfer (leave-one-out). Chance = 25% / 33% (d1–d3). d0 recall ≥89% inflates headline; d1–d3 mean: 45.9%. Bootstrap 95% CI half-widths ≤0.7 pp.

Holdout	Pythia Acc (%)	OLMo Acc (%)	GPT-2 XL Acc (%)
nucleus	**69.4**	**65.4**	**59.9**
temp (τ=0.9)	62.8	58.2	56.1
top-k (k=50)	51.5	54.9	53.6

balanced accuracy averages 55.4% (chance = 33%), with d2 dropping to 26.8% (Pythia holdout).

5.3 Cross-Strategy Transfer

Leave-one-strategy-out results appear in Table 7.

Mean cross-strategy accuracy (59.1%) falls 7pp below cross-model (66.4%); excluding d0, d1–d3 accuracy is 45.9% (chance = 33%). d2 recall varies sharply (nucleus holdouts 30–54%, top-k 8–19%, temperature 3.7–39.5%). Training on nucleus + top-k and testing on temperature inverts the d1/d2 boundary (d1v2 AUC < 0.5); per-strategy z-score normalization partially recovers it (Limitations). A multi-temperature ablation (Appendix A.6) confirms within-τ signal holds for $\tau \leq 0.9$ but collapses at τ=1.1; cross-τ LOO transfer (mean 46.7%) is harder than cross-strategy transfer.

End-to-end PLMs (DeBERTa-v3, RoBERTa) achieve only ∼59% (19pp below RF), confirming depth signal is inaccessible to standard PLM representations (Appendix A.8).

5.4 Preliminary Cross-Scale Observations

We extend to Qwen2.5-7B-Instruct (Team, 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) under 4-bit NF4 QLoRA (nucleus p=0.95, 500 samples/depth, depths 0–5). Both exceed chance: Llama-3.1-8B achieves 70.2% (95% CI $[68.3, 72.4]$, $\text{EDD}_{0.70}$=4) and Qwen2.5-7B 60.1% ($[58.0, 62.1]$, $\text{EDD}_{0.70}$>5), with weaker per-step signal (mean consecutive AUC ≈0.78 vs. 0.85 at 1–1.5B).

Base vs. Instruction-Tuned Control. A base-model control (Llama-3.1-8B) achieves 59.9% (95% CI [57.8, 62.15]) with non-monotonic AUC decay ($d_{2\text{v}3} = 0.682 \rightarrow d_{3\text{v}4} = 0.730$), while the instruction-tuned variant reaches 70.2% with strictly monotonic decay—suggesting instruction-tuning *regularizes* degradation patterns rather than introducing confounding artifacts.

Four confounds (quantization, reduced n, single strategy, vocabulary mismatch) limit causal attribution (Appendix A.12). Cross-model recursive chains (84.75%; Appendix A.13) and cross-domain transfer (arXiv 74.1%) further confirm signal robustness.

LoRA rank sensitivity. Depth signal persists across LoRA ranks 4–64 (accuracy spread 4.65 pp; Appendix A.15), confirming the phenomenon is not an artifact of a specific fine-tuning configuration.

6 Discussion and Conclusion

We formalized recursive generation depth estimation and showed that perplexity tail statistics achieve 70–85% four-class accuracy ($\text{EDD}_{0.75}$=4), with five features sufficing for 97.6% of full performance and classifier-agnostic convergence within ≤0.4pp (§4.4). The dominant features (`p90_ppl` top-2 in 9/9 cells, `var_surprisal` in 8/9) generalize across model families without retraining, making depth estimation a candidate early-warning signal for model collapse (Shumailov et al., 2024; Shi et al., 2025); surface-text ablation confirms the signal is inaccessible from text alone (42 pp below RF; §5.1). Cross-strategy transfer remains the primary bottleneck (59.1% vs. 66.4% cross-model), requiring a two-stage pipeline for deployment. Depth estimation complements binary detection (Xie et al., 2023; Wettig et al., 2024) and watermarking (Kirchenbauer et al., 2023) by enabling depth-graduated curation (Bevendorff et al., 2025) (pilot inconclusive; Appendix A.11).

Limitations

Cross-strategy deployment constraint. Cross-strategy transfer induces a d1/d2 boundary inversion under temperature sampling. Per-strategy z-score normalization partially corrects the inversion under logistic regression (GPT-2 XL d1v2 AUC 0.24→0.56, OLMo 0.19→0.70) but yields negligible improvement for the Random Forest

classifier used throughout the main experiments ($\Delta < 0.02$ pp), and even in the best case (logistic regression) leaves a residual gap relative to within-strategy performance ($\sim$0.83), indicating that a two-stage pipeline (strategy identification + strategy-conditioned classification) is needed for reliable cross-strategy deployment.

d0 inflation of headline metrics. Depth 0 (human-written text) is near-perfectly separable (recall >98% across all nine model–strategy settings), inflating headline accuracy. Cross-model transfer accuracy of 66.4% is substantially driven by d0; d2 recall drops to 26.8% in the worst case (Pythia holdout).

Controlled vs. ecological setting. The primary benchmark uses Wikipedia seed text with controlled recursive generation (LoRA fine-tuning, 1–1.5B parameter models); cross-scale experiments extend to 7–8B instruction-tuned models (§5.4). This protocol isolates the recursive depth variable but diverges from wild recursive contamination, where models are pretrained on mixed corpora containing prior-generation outputs at unknown proportions; the fingerprints we identify may attenuate or shift under such diffuse exposure. Depth signal is the dominant classification factor (5-fold CV 59–82%, up to 2.5× chance): LOGO accuracy (72.1%) matches standard CV (72.0%), and seed-group prediction is at chance (10.2% vs. 10%; Appendix A.9). Cross-scale experiments include both instruction-tuned and base variants; the base model shows weaker but above-chance depth signal (59.9% vs. 70.2%) with non-monotonic AUC decay, suggesting instruction-tuning regularizes degradation patterns. Real-world documents may also interleave segments at different recursive generation depths; extending to segment-level estimation introduces a granularity–statistical-power tradeoff.

Domain-dependent false positive rate. On editorial text (WikiText-103), all classifiers achieve ≥98.5% d0 accuracy, but on web-crawl data (C4, 5,000 passages), FPR ranges from 4.7% (OLMo) to 22.4% (GPT-2 XL). Misclassifications concentrate in template/SEO content with elevated repetition features (`rep_3gram` +37%, `rep_4gram` +75% vs. correctly classified d0); the model gap likely reflects training data composition (OLMo's Dolma includes diverse web data vs. Pythia/GPT-2 XL's Wikipedia/WebText). Deployment on heterogeneous web corpora requires model- and domain-specific calibration. A post-hoc repetition-ratio threshold could further reduce web-crawl false positives, which we leave to future work.

Adversarial robustness. Adversarial paraphrasing (Sadasivan et al., 2023) could disrupt the perplexity tail features that drive classification; depth estimation's robustness under such attacks is untested.

EDD monotonicity assumption. Our EDD definition (Eq. 1) requires all pairwise AUCs up to depth d^* to exceed T, which is well-behaved when AUC decays monotonically—as observed in all our configurations. However, Bakshi and Chakraborty (2026) show that early recursion can temporarily *improve* model quality, which could produce non-monotonic AUC profiles (e.g., a dip at d1/d2 followed by recovery at d2/d3). Our cross-strategy transfer results provide an empirical example: training on nucleus + top-k and testing on temperature yields d1-vs-d2 AUC as low as 0.129 (GPT-2 XL) and 0.253 (OLMo), violating the monotonicity assumption. In such cases EDD would conservatively report a lower value than the deepest discriminable pair, potentially underestimating the true discriminable range; extending EDD to handle non-contiguous discriminable regions is future work. The choice of threshold T is also application-dependent: at T=0.80, several cells lose one depth level relative to T=0.75.

Reference model dependency. Reference model choice affects classification accuracy by up to 11 pp; same-family references yield optimal performance, though all configurations exceed 2.5× chance. Surprisal-based features are computed relative to Pythia-1.4B, which coincides with one of the three generators. For the Pythia pipeline this constitutes self-model feature extraction, while OLMo and GPT-2 XL measurements are cross-model. Empirically, the self-model configuration does not yield the highest accuracy (Pythia 78.2% vs. GPT-2 XL 85.4%), suggesting the overlap does not confer a systematic advantage. An ablation with OLMo-1B and GPT-2 XL as alternative references shows accuracy ranging from 67–79% across three reference architectures (Appendix A.16), all exceeding 2.5× chance; same-family reference yields optimal performance ($\Delta \leq 11$ pp). The choice of reference model affects feature values; a different reference model may shift absolute feature magni-

tudes and potentially alter discriminability at depth boundaries.

Single-LoRA-instance limitation. All generated text at a given depth originates from a single LoRA instance per model, so seed-grouped cross-validation may overestimate generalization. However, leave-one-group-out evaluation (72.1%, matching 5-fold CV at 72.0%) and cross-model transfer (66.4%) partially mitigate this concern.

Additional caveats. Detection features degrade under domain shift (Pudasaini et al., 2026); our depth features are likely susceptible. Falahati et al. (2026) show that curated synthetic data need not trigger collapse, attenuating the fingerprints we exploit.

Ethics Statement

This work uses publicly available pretrained language models (Pythia-1.4B, OLMo-1B, GPT-2 XL) and Wikipedia text as seed data. No human subjects were involved, no private or sensitive data were collected, and no new models were released. Our method is intended to support data provenance and training-data auditing; we acknowledge that depth estimation tools could in principle be misused to circumvent content moderation, though such risks are shared with all detection-oriented research. Code and data will be released upon acceptance.

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A Additional Results

A.1 Top-k Feature Trajectories

Figure 2 in the main text shows feature trajectories for nucleus (p=0.95) and temperature (τ=0.9) sampling. Top-k (k=50) trajectories follow the same monotonically decreasing trend for all features but with compressed inter-depth spacing, consistent with the signal attenuation effect discussed in §4.5. Concretely, the d0→d1 drop in P90_PPL under top-k is 78% (vs. 87% for nucleus), and the d2→d3 reduction is 31% (vs. 39% for nucleus). Despite the magnitude reduction, feature rank ordering across depths is preserved, which explains why ordinal classifiers maintain directionality even under top-k conditions.

A.2 Permutation Importance

To verify that the Gini-based importance rankings in §4.2 are not an artifact of the impurity measure, we compute permutation importance (5-fold CV, 10 repeats) for all 9 atlas cells. The top-2 features are stable across measures: P90_PPL ranks in the top 2 in 9/9 cells under both Gini and permutation importance, and VAR_SURPRISAL in 8/9 (vs. 7/9 for Gini, with the additional cell being OLMo top-k; the single SHAP exception is GPT-2 XL top-k, where P75_PPL displaces VAR_SURPRISAL to rank 3). Kendall's W=0.90 across all three methods (Gini, permutation, SHAP) indicates strong concordance (p<0.005). Pairwise Spearman correlations are: SHAP–Gini ρ=0.93, SHAP–permutation ρ=0.84, Gini–permutation ρ=0.76. Per-cell top-5 overlap between SHAP and permutation importance is 3–4 out of 5 features; the aggregated (mean-across-cells) top-5 sets are identical, confirming that the top features are robust to the choice of importance measure at the aggregate level while per-cell rankings exhibit model-dependent variation.

A.3 Per-Class Recall

Table 8 reports per-class recall across all 9 atlas cells (4-class in-distribution, 5-fold CV). Depth 0 recall exceeds 98% uniformly. Among generated depths, d2 is the hardest class in every cell; the deficit is most pronounced under top-k sampling (46–53%), where vocabulary truncation removes the tail tokens that carry depth-discriminative signal.

Table 8: Per-class recall (%) for all 9 atlas cells. d2 is the lowest-recall class in every cell. Boldface marks the highest d2 recall.

Cell	d0	d1	d2	d3
Pythia, nuc.	99.7	81.9	58.7	72.3
Pythia, τ	99.7	81.3	53.9	72.1
Pythia, top-k	99.5	72.3	45.6	57.5
OLMo, nuc.	98.8	80.5	61.5	74.3
OLMo, τ	99.7	87.7	66.1	77.2
OLMo, top-k	99.5	78.5	51.2	65.6
GPT-2 XL, nuc.	99.2	85.9	**73.3**	77.5
GPT-2 XL, τ	99.5	90.9	68.8	81.9
GPT-2 XL, top-k	98.5	79.5	52.5	71.1

A.4 DCOL Iteration Details

Table 9 summarizes all six DCOL iterations. DeBERTa-v3-large exhibited systematic numerical instability across five configurations; RoBERTa-large resolved stability but could not close the accuracy gap with feature-based methods.

Table 9: DCOL iteration summary. All experiments use Pythia-1.4B nucleus data. RF baseline: 78.2%.

Ver.	Backbone	Config	Outcome
v1	DeBERTa	FT, m_lr 10^{-3}	Embed. collapse (31.5%)
v2	DeBERTa	FT, m_lr 10^{-4}	OOM
v3	DeBERTa	FT, m_lr 10^{-4}	Forgetting + OOM
v4	DeBERTa	LoRA, m_lr 10^{-3}	Embed. collapse
v5	DeBERTa	LoRA, m_lr 10^{-4}	NaN gradients
v6	RoBERTa	FT, grad clip 1.0	Stable; 64.5% (-13.7pp)

A.5 Zero-Shot LLM Prompt Template

Both DeepSeek V3.2 and Qwen2.5-7B-Instruct use identical prompting. The system message defines recursive generation depth with per-depth descriptions (depth 0: human-written; depth 1–3: successive LLM generation cycles) and lists expected textual patterns at each depth (vocabulary richness, repetition, coherence). The user message presents the passage (truncated to ≈384 words) and requests a JSON response containing an integer depth prediction (0–3), a confidence score, and a one-sentence reasoning. No in-context examples are provided.

> **System:** "You are an expert at detecting AI-generated text and estimating its generation depth. 'Recursive generation depth' refers to how many times a text has been through an LLM generation cycle: Depth 0: Original human-written text [...] Depth 3: Text generated by an LLM given depth-2 text as input. Key patterns by depth: Depth 0 (human): Rich vocabulary [...] Depth 3 (third-gen): Noticeable repetition, formulaic structure [...]"
>
> **User:** "Analyze the following text and estimate its recursive generation depth (0, 1, 2, or 3). TEXT: {text}. Respond with ONLY a JSON object: {"depth": ⟨int 0–3⟩, "confidence": ⟨float 0–1⟩, "reasoning": ⟨str⟩}"

A.6 Temperature Robustness

The cross-strategy results in §5.3 use a single temperature (τ=0.9). To test whether depth signal generalizes across temperatures, we repeat the classification pipeline at $\tau \in \{0.5, 0.7, 0.9, 1.1\}$ on Pythia-1.4B.

Within-τ robustness. For $\tau \in [0.5, 0.9]$, d1-vs-d2 AUC exceeds 0.80 (peak 0.851 at τ=0.7, comparable to nucleus p=0.95's 0.861) and 4-class accuracy ranges from 75.4% to 77.3% (Table 10), confirming the depth signal is not an artifact of a single temperature. The same top-5 features (`var_surprisal`, `p90_ppl`, `mean_ppl`, `mean_surprisal`, `p75_ppl`) dominate across all τ values with only minor rank-order variation.

Collapse at τ=1.1: memorization as prerequisite. At τ=1.1, 4-class accuracy drops to 57.2% and d1v2 AUC falls to 0.528. Training loss reveals the mechanism: at low temperature (τ=0.5), fine-tuning loss decreases steeply across recursive depths ($2.27 \rightarrow 0.42 \rightarrow 0.16$), indicating rapid memorization of the model's own output. At τ=1.1, loss remains nearly flat ($2.27 \rightarrow 1.91 \rightarrow 1.85$)—high-entropy output resists memorization, so depth-dependent distributional shifts do not accumulate. This establishes *memorization as a prerequisite* for detectable depth signal: recursive depth is discriminable only when successive fine-tuning iterations progressively reshape the output distribution.

Cross-τ non-transferability. Leave-one-τ-out transfer yields mean accuracy of 46.7% with mean d1v2 AUC of 0.319 (Table 11)—worse than cross-strategy transfer (59.1%; §5.3). Holdout τ=0.5 produces inverted d1v2 AUC (0.176), confirming that different temperatures produce surprisal distributions with incompatible shapes. This extends the strategy-specificity finding: even within a single decoding family, the feature space is temperature-specific, reinforcing the need for strategy-conditioned classifiers.

Ecological validity. On WikiText-103 (546 editorial passages), all three model-specific RF classifiers achieve ≥98.5% d0 accuracy despite +621% domain shift in mean perplexity—partly an easy case, as WikiText's higher perplexity places it farther from d1–d3 in feature space. On noisier web-crawl data (5,000 C4 passages), FPR ranges from 4.7% (OLMo) to 22.4% (GPT-2 XL), concentrated in template/SEO content with high n-gram repetition (Table 13); deployment on heterogeneous web data requires model-specific calibration.

Table 10: Within-τ 4-class classification (Pythia-1.4B, 5-fold CV). divj: pairwise AUC for depth i vs. depth j.

τ	4-class acc (%)	d1v2 AUC	d0v1 AUC	d2v3 AUC
0.5	77.3	0.842	0.996	0.764
0.7	75.4	0.851	0.997	0.720
0.9	75.9	0.808	0.991	0.781
1.1	57.2	0.528	0.988	0.599

Table 11: Cross-τ leave-one-out transfer (Pythia-1.4B). Train on three temperatures, test on the held-out one.

Holdout	Acc (%)	d1v2 AUC
τ=0.5	52.2	0.176
τ=0.7	41.0	0.299
τ=0.9	46.4	0.277
τ=1.1	47.1	0.525
Mean	46.7	0.319

A.7 Feature Overlap Analysis

We probe whether end-to-end PLM representations capture the same depth signals as our statistical features via linear probing of [CLS] representations from DeBERTa-v3-base and RoBERTa-base. PLM representations excel at surface lexical features (TTR: R^2=0.85, repeated bigrams) but fail to capture perplexity distribution statistics—VAR_SURPRISAL and P90_PPL yield negative R^2 under linear probing, indicating these features are not linearly accessible in the pretrained representation space. This confirms that the 19pp accuracy gap between feature-based and end-to-end methods (§A.8) stems from complementary—not redundant—representations: PLMs capture lexical patterns while the targeted statistical features capture distributional tail properties that constitute the dominant depth markers.

A.8 End-to-End Learning

DeBERTa-v3-base (He et al., 2023) and RoBERTa-base (Liu et al., 2019) finetuned end-to-end achieve 58.3% (±4.7%) and 59.5% (±3.2%) respectively—lagging the RF baseline (78.2%) by 19.9pp and 18.7pp. The 1.2pp inter-backbone gap rules out architecture bias; Table 12 shows both backbones share the same failure mode—high d0 F1 (>0.94) but collapsed d1/d2 performance (F1 ≤ 0.42)—while RF achieves 0.596 d2 recall, a >2× advantage at the hardest depth class.

Table 12: Per-class performance for end-to-end PLM classifiers vs. RF baseline (Pythia-1.4B nucleus). PLM columns report F1; RF d2 column reports recall (the class with the largest PLM–RF gap). Both PLM backbones share the same failure pattern: collapsed d1/d2 scores despite adequate d0/d3.

Method	Acc (%)	d0	d1	d2	d3
DeBERTa-v3	58.3 ± 4.7	.947	.378	.247	.656
RoBERTa-base	59.5 ± 3.2	.958	.416	.261	.647
RF (19 feat.)	78.2	—	—	.596†	—

†Recall; shown for the depth class with the largest PLM–RF gap.

Feature space divergence. SHAP TreeExplainer analysis confirms that perplexity and surprisal features account for 88.8% of RF's total contribution. Three independent importance measures—SHAP, Gini, and permutation—show strong concordance (Kendall's W=0.90, p<0.005; pairwise Spearman ρ: SHAP–Gini 0.93, SHAP–permutation 0.84; §A.2), with `p90_ppl` appearing in the SHAP top-2 in 9/9 cells and `var_surprisal` in 8/9. Linear probing of PLM [CLS] representations yields negative R^2 for both `var_surprisal` and `p90_ppl` (§A.7), indicating that pretrained representations lack the linear structure needed to recover perplexity tail statistics—a systematic representational gap rather than a backbone-specific artifact.

DCOL (Pitawela et al., 2025) reached 64.5% (§A.4); all end-to-end approaches (58–65%) underperform feature-based methods, suggesting depth

estimation benefits from targeted statistical features that capture distributional tail properties absent from PLM representations. This comparison is bounded by backbone scale ($\leq$125M parameters); whether larger PLMs (e.g., 7B+) can close the gap via richer internal representations remains an open question.

A.9 Domain Bias Analysis

To assess whether the classifier exploits depth-related features or memorizes stylistic cues from individual seed articles, we conduct three sub-experiments.

Methodological note. An initial implementation assigned sample IDs with a global counter across depths, inadvertently correlating seed-group identity with depth (LOGO accuracy 64.0%, seed-group prediction 20.8%). We discovered the issue during internal review, corrected to per-depth seed assignment ensuring each group contains equal samples per depth, and re-ran all sub-experiments. The results below use the corrected pipeline; the initial (buggy) numbers are reported for transparency.

Sub-A: Standard CV. Five-fold CV accuracy on 3-class (d1/d2/d3) classification ranges from 59% to 82% across the 9 atlas cells (up to 2.5$\times$ above the 33.3% chance level), confirming that depth signal is the dominant classification factor.

Sub-B: LOGO evaluation. We assign each passage to one of 10 seed-article groups (balanced: each group contains equal samples per depth) and perform leave-one-group-out CV. LOGO mean accuracy is 72.1% (range 59.5–81.6%), matching standard CV (72.0%) within noise (before fix: 64.0%, reflecting the depth–group confound). Per-cell fold standard deviations range from 0.4pp to 1.4pp, indicating that cross-group generalization is as reliable as within-group classification.

Sub-C: Negative control. We train two classifiers on identical features: one predicting depth, the other predicting seed-group identity (10-class, chance = 10%). Depth prediction accuracy averages 72.0%; seed-group prediction averages 10.2%—indistinguishable from chance (before fix: 20.8%, inflated by the depth–group confound). The depth-to-seed-group accuracy ratio is 7.1$\times$, confirming that the feature representation encodes depth but not seed-article identity.

Taken together, these results show that seed-article grouping has no measurable effect on depth estimation after correcting the group assignment: LOGO accuracy equals standard CV, and seed-group prediction is at chance.

A.10 Wild-Data Calibration

Table 13 reports d0 accuracy and false positive rate (FPR) for all three model-specific nucleus RF classifiers on two out-of-distribution human-text corpora: WikiText-103 (546 editorial passages) and C4 (5,000 web-crawl passages).

Table 13: Wild-data d0 accuracy and FPR for nucleus RF classifiers. All passages are human-written (true d0). FPR = fraction misclassified as d1+. 95% Wilson CIs computed for C4 (n=5,000).

Model	Source	n	d0 (%)	FPR (%)	95% Wilson CI
Pythia-1.4B	WikiText	546	100.0	0.0	—
	C4	5,000	79.8	20.2	[19.1, 21.3]
GPT-2 XL	WikiText	546	99.3	0.7	—
	C4	5,000	77.6	22.4	[21.3, 23.6]
OLMo-1B	WikiText	546	98.5	1.5	—
	C4	5,000	95.3	4.7	[4.2, 5.3]

On WikiText-103, all three classifiers achieve $\geq$98.5% d0 accuracy, confirming that editorial text is reliably distinguished from recursive generations across model families. On C4, FPR varies sharply: OLMo (4.7%) vs. Pythia (20.2%) and GPT-2 XL (22.4%). C4 misclassifications concentrate in template and SEO content with elevated repetition features (`rep_3gram` +37%, `rep_4gram` +75% vs. correctly classified d0 passages), rather than in random noise. The model gap likely reflects training data composition: OLMo (trained on Dolma, which includes diverse web data) produces d0 features more tolerant of web-crawl variation, while Pythia and GPT-2 XL (trained primarily on Wikipedia/WebText) map template-heavy content into the d1+ region. Among C4 false positives for GPT-2 XL, predictions distribute as d1 (9.0%), d2 (3.7%), d3 (9.6%); for OLMo, d1 (2.9%), d2 (0.5%), d3 (1.3%)—the non-uniform depth distribution suggests the classifier maps specific repetition patterns to particular depth signatures rather than producing random misclassifications.

A.11 Downstream Curation PoC

To validate that depth estimation translates to actionable data curation, we conduct a proof-of-concept downstream evaluation. We fine-tune Pythia-410M with LoRA (r=16, α=32) for 3 epochs on 8 data conditions derived from the Pythia

nucleus depth-estimation corpus, repeating each condition over 5 random seeds. Conditions vary the selection mechanism (RF-predicted depth, oracle depth, random subsampling) and stringency (strict ≈25%, moderate ≈50%, soft threshold). Table 14 reports per-condition perplexity; Table 15 reports pairwise contrasts.

Table 14: Curation PoC: per-condition perplexity (Pythia-410M LoRA, 5 seeds). Δ is signed so that positive values indicate lower (better) perplexity than unfiltered. * : $p < 0.05$ (paired t-test vs. unfiltered).

Condition	Mean PPL	Std	n	Δ vs Unfilt.	p	d_z
unfiltered	15.776	0.066	10,000	—	—	—
rf_soft	15.733	0.038	4,422	+0.043	0.407	0.41
rf_moderate	15.756	0.056	4,940	+0.021	0.492	0.34
rf_strict	15.810	0.061	2,496	−0.034	0.607	−0.25
oracle_moderate	15.809	0.103	5,000	−0.033	0.485	−0.34
oracle_strict	15.879	0.128	2,500	−0.103	0.216	−0.66
random_soft	16.062	0.079	4,422	−0.286	<0.001*	−2.77
random_moderate	15.998	0.090	4,940	−0.222	0.028*	−1.50
random_strict	16.239	0.165	2,496	−0.463	0.013*	−1.89

Table 15: Pairwise comparisons (paired t-test, 5 seeds). * : $p < 0.05$; ** : $p < 0.01$.

Comparison	Δ PPL	t	p
rf_strict vs random_strict	0.429	−7.05	0.002**
rf_moderate vs random_moderate	0.242	−3.77	0.020*
rf_soft vs random_soft	0.328	−17.98	<0.001**
rf_soft vs unfiltered	0.043	0.93	0.407

A.12 Cross-Scale Full Results

This section provides the full cross-scale analysis summarized in §5.4. We extend the benchmark to two 7–8B models: Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, both under 4-bit NF4 QLoRA fine-tuning, generating depths $d \in \{0, 1, \ldots, 5\}$ with nucleus sampling (p=0.95, 500 samples per depth). Feature extraction uses the same reference model (Pythia-1.4B) and 19-feature pipeline as all prior experiments.

Scale comparison. Table 16 contrasts the two 7–8B models with the three 1–1.5B models on matched evaluation (nucleus sampling, depths 0–3).

(1) Depth signal persists at 7–8B scale. Both 7–8B models substantially exceed the 25% chance baseline: Llama-3.1-8B achieves 70.2% (95% CI $[68.3, 72.4]$) and Qwen2.5-7B achieves 60.1% ($[58.0, 62.1]$), confirming that recursive depth remains estimable from perplexity-based features even when the generator is 5–7× larger than the reference model. Depth-0 separability is preserved in both models: d0v1 AUC is 0.949 (Llama-3.1) and 0.969 (Qwen2.5).

Table 16: Cross-scale comparison: 4-class RF accuracy (%, depths 0–3, nucleus), $\mathrm{EDD}_{T=0.70}$ (depths 0–5), and d0-vs-d1 AUC. CI: 95% bootstrap. Chance = 25%.

Model	4-class acc	$\mathrm{EDD}_{T=0.70}$	d0v1 AUC
GPT-2 XL (1.5B)	83.3	4	0.999
OLMo-1B	78.8	4	0.997
Pythia-1.4B	78.2	4	0.998
Llama-3.1-8B	70.2 [68.3, 72.4]	4	0.949
Qwen2.5-7B	60.1 [58.0, 62.1]	>5	0.969

(2) EDD at 7–8B scale. At threshold T=0.70, Qwen2.5-7B maintains all consecutive-pair AUCs above 0.70 through depth 5 (Table 17), yielding $\mathrm{EDD}_{T=0.70} > 5$—exceeding the EDD=4 observed at 1–1.5B. Llama-3.1-8B matches the smaller models at EDD=4: its monotonic decay causes d4v5 (0.618) to drop below T, despite stronger signal at shallower depths. Both larger generators produce weaker per-step signal than the 1–1.5B models (mean consecutive AUC: 0.786 for Llama-3.1, 0.782 for Qwen2.5, vs. 0.852 at 1–1.5B).

Table 17: Consecutive binary AUC for 7–8B models (nucleus p=0.95). Llama-3.1-8B exhibits monotonic decay (d4v5 drops below T=0.70); Qwen2.5-7B shows a non-monotonic (U-shaped) pattern with all pairs above T.

Model	d0v1	d1v2	d2v3	d3v4	d4v5
Llama-3.1-8B	0.949	0.877	0.771	0.716	0.618
Qwen2.5-7B	0.969	0.729	0.709	0.732	0.772

(3) Divergent AUC trajectories and instruction-tuning effect. Among instruction-tuned models, Llama-3.1-8B exhibits monotonic AUC decay consistent with the 1–1.5B pattern, while Qwen2.5-7B shows a U-shaped recovery beyond d2v3 (Table 17). A base-model control (Llama-3.1-8B without instruction tuning) clarifies the causal direction: the base model achieves 59.9% four-class accuracy (95% CI [57.8, 62.15], $\mathrm{EDD}_{0.70}$=2) with non-monotonic AUC decay (Table 18), while the instruction-tuned variant reaches 70.2% with strictly monotonic decay. The base-model control experiment suggests instruction-tuning regularizes rather than confounds the depth signal.

The base model exhibits non-monotonic decay with a U-shaped recovery at $d_{3\mathrm{v}4}$, while the

Table 18: Pairwise AUC: Llama-3.1-8B Base vs. Instruct.

Variant	d_{0v1}	d_{1v2}	d_{2v3}	d_{3v4}	d_{4v5}
Base	0.918	0.794	0.682	0.730	0.595
Instruct	0.965	0.874	0.771	0.716	0.618

instruction-tuned variant shows strictly monotonic decay. Both variants exceed chance (AUC > 0.5) at all boundaries, confirming depth signal exists regardless of instruction-tuning, though instruction-tuning strengthens and regularizes it.

Caveats. For Qwen2.5-7B, the CI lower bound (58.0%) does not exclude sub-60% accuracy; Llama-3.1-8B's lower bound (68.3%) is more comfortably above chance but still below the 75–83% range at 1–1.5B. The divergent AUC trajectories across two models caution against strong generalizations from only two 7–8B data points: both the monotonic (Llama-3.1) and non-monotonic (Qwen2.5) patterns are single observations, and further architectures are needed to establish which is the dominant regime at this scale.

Confounds. Four confounds limit these results: (1) 4-bit NF4 QLoRA quantization (absent at 1–1.5B) may distort the perplexity distribution independently of depth; (2) smaller sample size (500 vs. 5,000 per depth) widens confidence intervals; (3) evaluation is limited to nucleus sampling, leaving other strategies untested at this scale; (4) vocabulary mismatch between 7–8B generators and the 1.4B reference model. The base-model control partially addresses a former confound (instruction-tuning): since the base variant shows weaker signal with non-monotonic decay while the instruction-tuned variant shows stronger monotonic signal, instruction-tuning regularizes rather than confounds depth estimation. The remaining confounds are not mutually independent—quantization and vocabulary mismatch both affect the perplexity distribution—making it infeasible to attribute the observed accuracy gap (60–70% vs. 78–85% at 1–1.5B) to any single factor.

A.13 Cross-Boundary Generalization

The preceding experiments assume a single model family generates all recursive depths. In practice, recursive contamination may traverse model boundaries: text generated by model A is used to train model B, whose output trains model C. We probe whether the depth signal persists under two relaxations of the single-model assumption.

Cross-model recursive chains. We construct a heterogeneous chain: human text (d0) $\rightarrow$ Pythia-1.4B generates d1 $\rightarrow$ OLMo-1B finetuned on Pythia-d1 generates d2 $\rightarrow$ GPT-2 XL finetuned on OLMo-d2 generates d3. Each recursive step uses a *different* model family, eliminating the possibility that depth signal arises from a single model's idiosyncratic behavior. A 4-class Random Forest achieves 84.75% accuracy (95% CI $[81.0, 88.25]$), with per-depth F1 scores of 0.98 (d0), 0.88 (d1), 0.76 (d2), and 0.76 (d3). Pairwise AUC remains high: d0-vs-d1 = 0.999 and d2-vs-d3 = 0.841. These results suggest that distributional artifacts of recursive generation persist across model boundaries—providing preliminary evidence consistent with a process-level depth signal rather than individual generator artifacts.

Confound discussion. The heterogeneous chain confounds architecture identity with depth (d1=Pythia, d2=OLMo, d3=GPT-2 XL). However, if the classifier exploited architecture fingerprints rather than depth signal, near-perfect accuracy would be expected since architecture perfectly determines depth label. The observed 84.75% with monotonically decreasing per-depth F1 (d0=0.98, d1=0.88, d2=0.76, d3=0.76) is more consistent with a depth-signal interpretation, as architecture fingerprinting would not produce this graded pattern. We address this confound directly with a reverse-chain control below.

Reverse-chain control. To empirically test the architecture confound hypothesis, we constructed a reverse chain (GPT-2 XL $\rightarrow$ OLMo $\rightarrow$ Pythia) reversing the model-depth assignment. The reverse chain achieves 85.75% (95% CI [82.0, 89.25]), statistically indistinguishable from the forward chain (84.75%, [81.0, 88.25]). If the classifier exploited architecture fingerprints rather than depth signal, reversing the chain should invert depth predictions; the equivalent accuracy under both orderings provides empirical evidence against the architecture confound.

Cross-domain transfer. All prior experiments use WikiText-103 as the seed corpus. To test domain dependence, we repeat the pipeline with arXiv abstracts as the seed text (Pythia-1.4B, nucleus p=0.95, 500 samples per depth, d0–d3). Five-fold cross-validation yields 74.1% $\pm$ 2.0% accuracy,

compared to ∼78% on the WikiText seed. Per-depth F1 shows the familiar pattern: d0 = 1.00, d1 = 0.78, d2 = 0.52, d3 = 0.66—with d2 remaining the most challenging class. The 4pp accuracy gap relative to WikiText likely reflects the narrower stylistic range of scientific abstracts, which may reduce the distributional contrast between adjacent depths. These results provide preliminary evidence that the perplexity-based features capture properties of the recursive generation process rather than domain-specific artifacts of the seed corpus.

A.14 Extended Discussion

Depth-graduated curation. Current data curation operates on a binary human-vs-machine distinction (Xie et al., 2023; Wettig et al., 2024). Recursive generation depth estimation enables a finer-grained alternative: retain low-depth generations (d1) whose distributional properties remain close to the human source, while down-weighting or filtering high-depth text (d3+) where tail erosion has degraded quality. The 70–85% in-distribution accuracy and $\mathrm{EDD}_{0.75}{=}4$ suggest depth-graduated filtering is feasible within known model–strategy configurations, though the d1/d2 boundary inversion under cross-strategy transfer (§5.3) means reliable curation requires either known decoding strategy metadata or an upstream strategy-identification step; a pilot study (Appendix A.11) yields an inconclusive RF-vs-unfiltered comparison ($p > 0.4$), though RF-filtered subsets significantly outperform size-matched random subsampling ($p < 0.001$–0.02); large-scale validation remains open. Importance weighting (Sugiyama et al., 2007) could further enable continuous depth-aware data selection rather than discrete thresholding.

Toward a provenance toolkit. Depth estimation complements existing provenance methods along orthogonal axes. Watermarking (Kirchenbauer et al., 2023) provides high-confidence attribution when the generator cooperates but is inapplicable to unwatermarked text. Source attribution (Uchendu et al., 2020) identifies the generating model but not the generation history. Depth estimation characterizes *iterative reuse* regardless of generator cooperation or identity. A layered provenance system combining these three signals—watermark presence, source model, and recursive generation depth—could support more nuanced data auditing than any single method.

A.15 LoRA Rank Sensitivity

Table 19: LoRA rank sensitivity on Pythia-1.4B nucleus (p=0.95). Depth signal is robust across ranks (spread 4.65 pp).

Rank	4-class Acc (%)	d1v2 AUC
4	71.8	0.790
16	76.5	0.843
64	73.1	0.833

A.16 Reference Model Ablation

Table 20: Reference model ablation on Pythia-1.4B nucleus data. All configurations exceed 2.5× chance (25%).

Reference Model	4-class Acc (%)	d1v2 AUC	Δ vs P
Pythia-1.4B (matched)	78.5	0.868	—
OLMo-1B	74.5	—	−4.0
GPT-2 XL (cross-arch)	67.3	0.727	−11.2